

A Simple AUTOMATIC BATHROOM LIGHT

SURESH DWIVEDI

An automatic bathroom light can be made in several ways, depending on your preference and budget. Some of the options make use of either a motion sensor, a smart switch, an occupancy sensor, or a photo sensor. The simple system proposed here makes use of a small magnet fixed in the bathroom door's frame and a reed switch with normally open contacts as the sensor on the bathroom's door, just opposite the magnet. So, whenever you open the door, the reed switch will move away from the magnet.

Be sure to consult a qualified electrician if you're not comfortable with electrical work or if you need

help with installation. Safety should always be your top priority when working with electricity.

The author's prototype is shown in Fig. 1 and the circuit diagram is shown in Fig. 2.

The circuit comprises step-down transformer X1, bridge rectifier BR1, 5V voltage regulator LM7805 (IC1), a normally closed type of reed switch, a small magnet, transistor BC557 (T1), op-amp 741 (IC2), decade counter 4017 (IC3), transistor 2N2219 (T2), 5V single changeover relay RL1, and a few other components. The circuit needs 5V DC to work, which can be derived from the 230V AC primary to 9V, 500mA

step-down transformer X1. The 9V AC from X1 goes to bridge rectifier BR1 for rectification and is filtered by capacitor C1. The rectified and filtered voltage is regulated to 5V DC by IC LM7805, which is indicated by the glowing of LED1.

When the bathroom door is closed, the magnet and the reed switch are near each other, reed switch contacts are open, and BC557 transistor T1 does not conduct. Hence voltage at pin 3 of IC2 remains lower than the reference voltage at pin 2 (around half of the supply voltage), output of op-amp IC2 at its pin 1 remains low, and decade counter 4017 (IC3) does not get any clock signal.

When the door is opened, the magnet and the reed switch move away from each other, the reed switch contacts close, and transistor T1 conducts. Hence voltage at pin 3 of IC2 goes higher than the reference voltage at pin 2, output of op-amp IC2 at its pin 1 goes from low to high, and decade counter 4017 (IC3) gets a clock signal. Clock Q0 of 4017 shifts to Q1 output and relay driver transi-

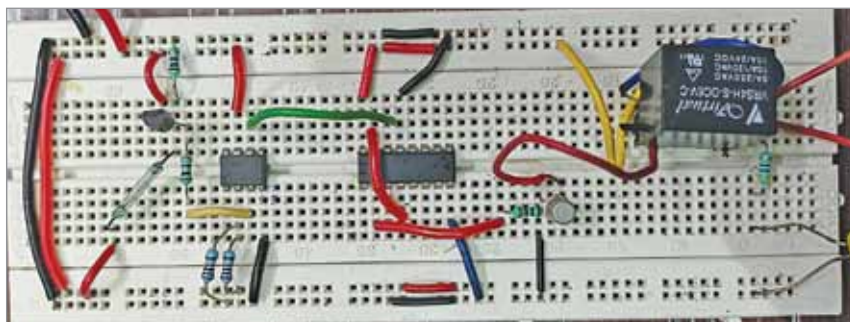


Fig. 1: Author's prototype

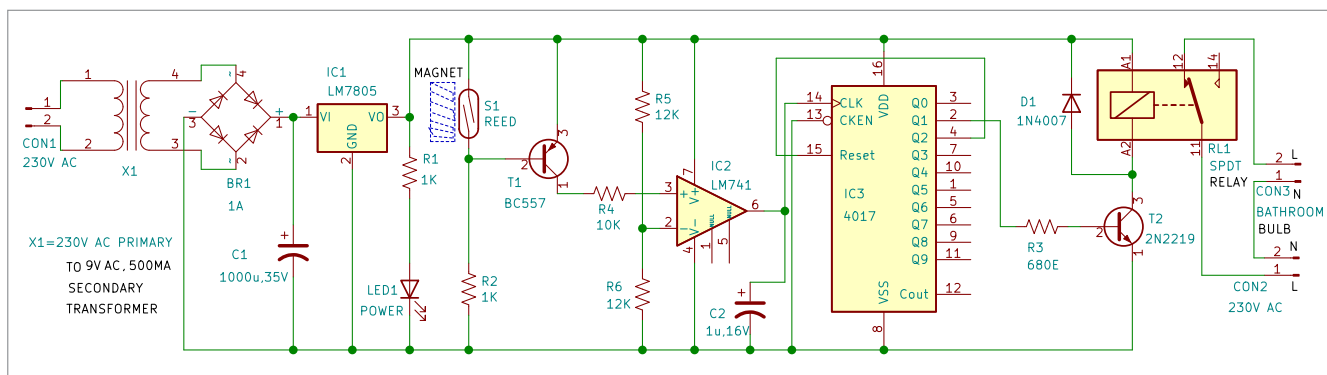


Fig. 2: Circuit diagram